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Introduction

"Whatever you would make habitual, practice it; and if you would not make a thing habitual, do not practice it, but accustom yourself to something else." – *Epictetus*

This book is a companion to a course in business statistics. It teaches the core ideas of statistics and how to apply them in Excel. Each chapter introduces a few concepts, works through examples, and ends with practice problems. The material takes effort, but steady work will get you through it.

I owe thanks to my business statistics students, especially Haipei Liu, Addison Alvine, and Sara Blair, for their feedback and careful reviews. Any remaining errors are mine.

Learning from Data

Statistics is about learning from the world. We want to know how customers behave, whether a new process works, or where a market is headed, but we can never observe everything. Instead, we collect **data**, a small and imperfect view of what we care about, and use it to draw conclusions about the bigger picture.

That last step comes with a catch. A sample never perfectly mirrors the population it came from, so two analysts with different samples will get different answers. **Probability** lets us measure how far our data might mislead us. With it, we can test a **hypothesis**, an idea we hold about the world, and decide whether the evidence supports it or whether we are just seeing noise.

You already reason this way in everyday life. You form impressions from limited experience, update them when new evidence arrives, and try not to be fooled by coincidence. Statistics teaches you to do the same thing carefully.

Why Excel?

Excel is the most widely used data tool in business. Finance, accounting, marketing, and operations teams use it every day to organize data, find patterns, and report results. It is available almost everywhere, and the skills you learn here will be useful in your first internship.

Excel can handle every technique in this book, from descriptive statistics and pivot tables to probability, regression, and hypothesis testing. Each chapter shows you how, with step-by-step instructions, screenshots, and downloadable datasets. You do not need any programming experience.

Excel in the Era of AI?

AI tools can now run an analysis in seconds, but you still need to understand the data and the logic to know whether the results make sense. Working in Excel builds that understanding. You learn to organize information, check your assumptions, and follow a calculation from start to finish. These habits make you a better user of AI, because you will be able to question its output instead of simply accepting it.

Roadmap

The book follows the same path as learning from data. **Descriptive Statistics** shows you how to summarize, visualize, and explore a data set. **Regression** introduces tools for measuring relationships between variables and making predictions. **Probability** gives you a way to model the uncertainty that comes from working with samples. **Inference** brings these pieces together so you can reason from a sample to the population and test hypotheses. The final chapter, **Regression and Inference**, uses those tools to judge how reliable a regression model is.

If you are new to spreadsheets, or just want a quick reference, start with the **Excel Basics** chapter at the end of the book. It covers how Excel organizes data (cells, ranges, worksheets, and the Data Analysis ToolPak) and lists every function used in the book.

How to Use This Book

Every chapter has the same structure. It starts with key concepts, formulas, and worked examples. Next, it shows how to carry out each technique in Excel, followed by an **Excel Function Summary**. Many chapters include a dataset you can download to follow along, and the online version adds **Try it** boxes: interactive graphs you can adjust to see a concept in action. Each chapter ends with **Exercises**. Try each problem on your own before clicking to reveal the answer.

How do I get Excel?

William & Mary students can download Office 365, which includes Excel, at https://software.wm.edu/office_365_d1/.

Part I

Describe

1 Descriptive Stats I

Understanding the nature and classification of data is crucial for effective analysis and decision-making. Data are the building blocks of insights, providing a foundation for businesses, researchers, and policymakers to make informed choices. How we collect and categorize data significantly impacts how it is analyzed and interpreted. Below I highlight key types of data and their unique characteristics to help you get started with data analysis. In the arc of this book, this is where learning from data begins: before we can summarize what we have observed or infer what it means for the world, we first have to understand the raw material itself — what kind of data we have collected and how it is organized.

1.1 Data and Types of Data

Let's start by defining data and the types of data structures that you might encounter when conducting analysis. **Data** are facts and figures collected, analyzed and summarized for presentation and interpretation. Data can be classified as:

- **Cross Sectional Data** refers to data collected at the same (or approximately the same) point in time. *Ex: NFL standings in 1980 or Country GDP in 2015.*
- **Time Series Data** refers to data collected over several time periods. *Ex: U.S. inflation rate from 2000-2010 or Tesla deliveries from 2016-2022.*
- **Structured Data** resides in a predefined row-column format (tidy). *Ex: spreadsheet data.*
- **Unstructured Data** does not conform to a pre-defined row-column format. *Ex: Text, video, and other multimedia.*

EXAMPLE

Consider a retail store analyzing its sales performance. If the store collects data on the total revenue generated *by each location* on Black Friday, it is cross-sectional data. On the other hand, if the store tracks *weekly sales for the past year* to observe trends, it is time series data. Structured data, like sales figures stored in *spreadsheets*, allows for easy comparison and analysis. Meanwhile, customer feedback gathered from *social media posts and video reviews* represents unstructured data, requiring advanced tools to extract meaningful insights.

1.2 Data Sets

A **data set** contains all data collected for a particular study. Understanding the structure the data set aids in communicating findings to stakeholders clearly and effectively. Data sets are composed of:

- **Elements** are the entities on which data are collected. *Ex: Football teams, countries, and individuals.*
- **Variables** are a set of characteristics collected for each element. *Ex: Goals scored, GDP, weight.*
- **Observations** are the set of measurements obtained for a particular element. *Ex: Salah, 20 (goals), 15 (assists). US, 2.3 (inflation), 4.5% (federal interest rate).*

Elements	Variable 1	Variable 2
Element 1	#	#
Element 2	#	#
Element 3	#	#
...	...	...

EXAMPLE

Consider the dataset on electric vehicles (EVs) displayed below:

Make	Model	Range_km	TopSpeed_kmh	Price_pounds	Charge_kmh
Tesla	Model 3	415	201	39,990	690
Tesla	Model S	652	250	79,990	750
Nissan	Leaf	270	144	26,995	100
BMW	i3	260	150	37,955	100
Audi	e-tron	400	200	60,000	470
Ford	Mustang	483	180	42,995	750
...	...	...	...	...	...

In this data set, each row represents an electric vehicle model, making the elements the specific EV models rather than the manufacturers. The variables collected for each model include:

- Make: The manufacturer of the EV.
- Model: The specific name of the EV model.

- Range_km: Driving range in kilometers on a full charge.
- TopSpeed_kmh: Maximum speed in km/h.
- Price_pounds: Price in pounds (£).
- Charge_kmh: Charging speed in kilometers per hour.

An example observation is "Tesla Model 3," with the following data: Make: Tesla, Model: Model 3, Range_km: 415, TopSpeed_kmh: 201, Price_pounds: 39,990, Charge_kmh: 690.

1.3 Scales of Measurement

Understanding scales of measurement is crucial for analyzing and interpreting data effectively in business. By distinguishing between categorical (e.g., marital status, satisfaction ratings) and numerical data (e.g., profits, prices), you'll know what methods to use for analysis. Knowing whether data is nominal, ordinal, interval, or ratio ensures your analysis and conclusions are accurate and relevant.

The **scales of measurements** determine the amount and type of information contained in each variable. In general, variables can be classified as **categorical** or **numerical**.

- **Categorical** (qualitative) data includes labels or names to identify an attribute of each element. Categorical data can be **nominal** or **ordinal**.
  - With **nominal** data, the order of the categories is arbitrary. Numbers (if used) serve only as identifiers. Common statistics used are based on counting frequencies and the mode (most common category) *Ex: Marital Status, Race/Ethnicity, or NFL division.*
  - With **ordinal** data, the order or rank of the categories is meaningful, but intervals between ranks are not necessarily equal or quantifiable. Common statistics are based on median, percentiles, or rank-order correlations. *Ex: Rating, Difficulty Level, or Spice Level.*
- **Numerical** (quantitative) include numerical values that indicate how many (discrete) or how much (continuous). The data can be either **interval** or **ratio**.
  - With **interval** data, the distance between values is expressed in terms of a fixed unit of measure. The zero value is arbitrary and does not represent the absence of the characteristic. Ratios are not meaningful. Common statistics are the arithmetic mean, standard deviation, or the Pearson correlation, *Ex: Temperature or Dates.*
  - With **ratio** data, the ratio between values is meaningful. The zero value is not arbitrary and represents the absence of the characteristic. All statistics are permissable including geometric mean, and the coefficient of variation. *Ex: Prices, Profits, Wins.*

EXAMPLE

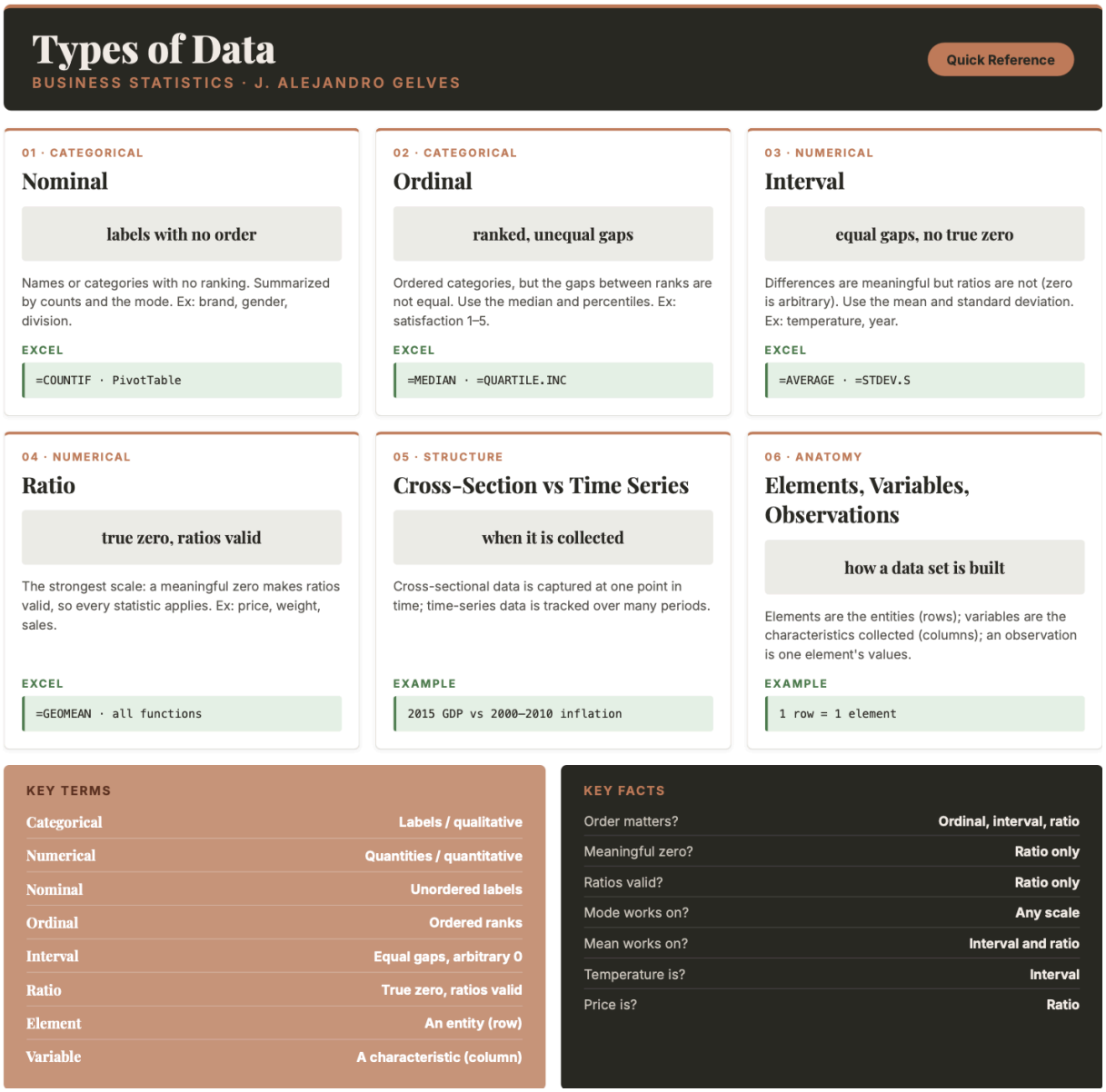
Let's keep using the EV example. Consider the new data set below:

Car	Brand	Range	Rating	Year
Model 3	Tesla	415	4	2021
Model S	Tesla	652	5	2021
Leaf	Nissan	270	3	2021
i3	BMW	260	3	2021
e-tron	Audi	400	4	2021
Mustang	Ford	483	4	2021
...	...	...	...	...

The variables can be classified as follows:

- Car (Categorical - Nominal), consists of names of cars, which are labels used to identify each row. The order of these names does not matter, making it nominal data.
- Brand (Categorical - Nominal) represents the manufacturer of the car (e.g., Ford, Audi). These are labels with no inherent order, making it nominal data.
- Range (Numerical - Ratio), refers to the car's driving range in miles. It is numerical and ratio because it has a meaningful zero (a car with zero range cannot move), and ratios are meaningful (e.g., a car with 250 miles range has double the range of one with 125 miles).
- Rating (Categorical - Ordinal) represents a rank or score (e.g., 4, 3, 2). The order matters, as higher ratings indicate better performance. However, the intervals between ratings are not consistent, so it is ordinal data.
- Year (Numerical - Interval) represents a point in time. While numerical, it is interval data because the zero point is arbitrary (e.g., year 0 does not indicate the "absence" of time), and ratios are not meaningful (e.g., 2020 is not "twice as late" as 1010).

1.4 Chapter Summary Cheat Sheet



1.5 Exercises

The following exercises will help you test your knowledge on the Scales of Measurement. In these exercises you will:

- Identify numerical and categorical data.
- Classify data according to their scale of measurement.

Answers are in the Answers to Exercises at the end of the book. Try not to peek until you formulate your own answer and double checked your work for any mistakes.

Exercise 1

A bookstore has compiled data set on their current inventory. A portion of the data is shown below:

Title	Price	Year Published	Rating
Frankenstein	5.49	1818	4.2
Dracula	7.60	1897	4.0
...	...	...	...
Sleepy Hollow	6.95	1820	3.8

1. Which of the above variables are categorical and which are numerical?

Answer on page ?? →

2. What is the measurement scale of each of the above variable?

Answer on page ?? →

Exercise 2

A car company tracks the number of deliveries every quarter. A portion of the data is shown below:

Year	Quarter	Deliveries
2016	1	14800
2016	2	14400
...	...	...
2022	3	343840

1. What is the measurement scale of the Year variable? What are the strengths and weaknesses of this type of measurement scale?

Answer on page ?? →

2. What is the measurement scale for the Quarter variable? What is the weakness of this type of measurement scale?

Answer on page ?? →

3. What is the measurement scale for the Deliveries variable? What are the strengths of this type of measurement scale?

Answer on page ?? →

Exercise 3

A coffee shop chain collects customer feedback data for analysis. For each visit/transaction recorded in their database, the following variables are stored:

- Customer ID: A unique identifier assigned to each registered customer (e.g., C001234, C005678).
- Satisfaction Level: Customers rate their overall experience as 1 (Very Dissatisfied), 2 (Dissatisfied), 3 (Neutral), 4 (Satisfied), or 5 (Very Satisfied).
- Visit Duration: The time spent in the store in minutes (e.g., 15, 42, 8).
- Beverage Type: The main drink ordered (e.g., Latte, Cappuccino, Americano, Tea, Other).

Answer on page ?? →

2 Descriptive Stats II

Understanding and visualizing data distributions is a fundamental step in data analysis. It provides critical insights into the underlying characteristics of the data, which directly impact decision-making, model performance, and interpretation. Below, we introduce tabular and visual techniques to describe your data. This is the *describe* stage of learning from data: we organize the raw observations into tables and pictures so their patterns become visible, before attempting to draw any conclusions from them.

2.1 Frequency Distributions (Categorical)

A **frequency distribution** is perhaps the most valuable tool for summarizing categorical data. It illustrates with a table the number of items within distinct, non-overlapping categories. An alternative known as the **relative frequency** quantifies the proportion of items in each category relative to the total number of observations. You can calculate it by taking the frequency of a particular class (f_i), and dividing it by the number of observations n. Relative frequency helps contextualize the data by highlighting the significance of each category compared to the whole.

The frequency distribution can be illustrated by using a bar plot. A **bar plot** illustrates the frequency distribution (or relative frequency) of categorical data. It includes the classes in the horizontal axis and frequencies or relative frequencies in the vertical axis and has gaps between each bar.

EXAMPLE

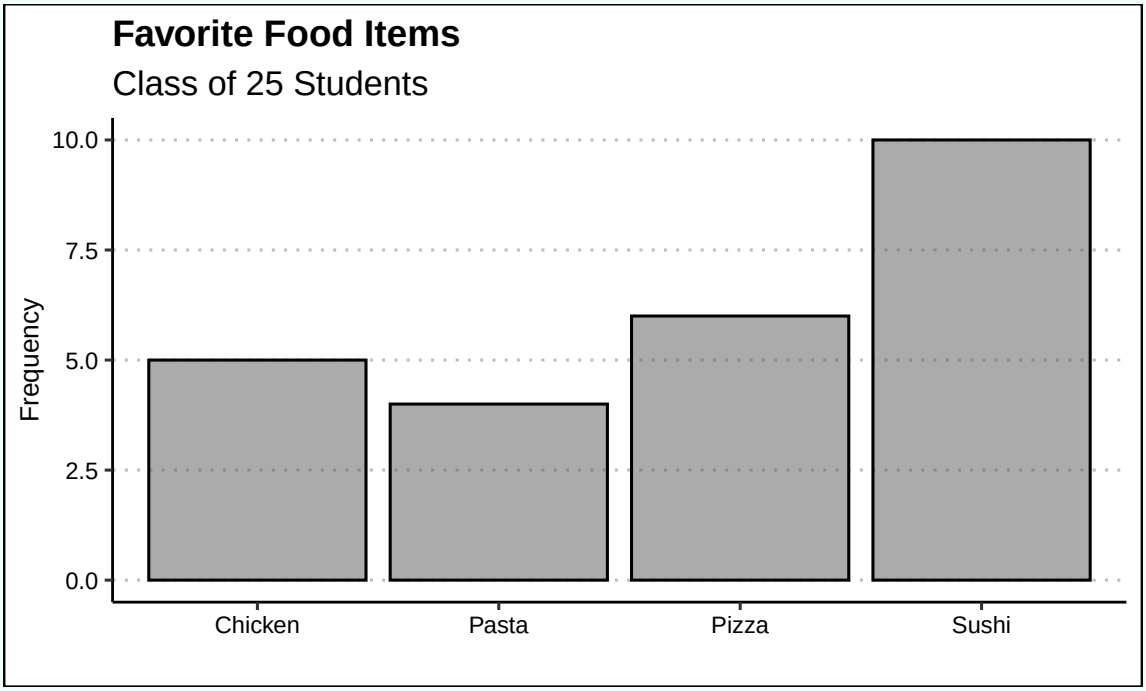
Consider data on students' answers to the question, what is your favorite food? You can see the data below:

PIZZA	SUSHI	SUSHI	CHICKEN	CHICKEN
PASTA	PASTA	PASTA	SUSHI	PASTA
CHICKEN	PIZZA	CHICKEN	SUSHI	PIZZA
SUSHI	SUSHI	SUSHI	SUSHI	PIZZA
PIZZA	CHICKEN	SUSHI	PIZZA	SUSHI

Simply observing raw data can make identifying the most and least popular items challenging. A frequency distribution organizes this information into a clear table, showcasing the popularity of each item. The frequency distribution of the table is displayed below:

Food	Frequency	Relative
Chicken	5	0.20
Pasta	4	0.16
Pizza	6	0.24
Sushi	10	0.40

Each food item is tallied up, and the result is shown in the frequency column. Alternately, we can show the tally as a proportion of the total (i.e., 25). For example, five students liked chicken; out of the 25 students surveyed, this represents 0.2 or 20%. This calculation is shown for each food item in the relative frequency column.
Below, you can see the bar graph showing the frequency distribution of the food items data. Note that the visualization is constructed by showing each food item as a bar with a height equal to the frequency.



2.2 Frequency Distributions (Numerical)

When working with numerical data, building a frequency distributions requires additional steps compared to categorical data. The challenge lies in the absence of predefined categories or classes. To construct a frequency distribution for numerical data, it is essential to determine the

number, width, and limits of the classes. Here are the steps to create a frequency distribution when data is numerical:

1. Determine the Number of Classes: The number of classes can be estimated using the 2^k rule, where k is the smallest integer such that 2^k exceeds the total number of observations by the least amount. This ensures the chosen number of classes provides a reasonable level of granularity for summarizing the data.

EXAMPLE

If a data set has 50 observations, we would choose six classes since $2^6 = 64$ is greater than 50 by the least amount.

Alternatively, you can use **Sturges' rule** to estimate the number of classes. This rule suggests using $k = \lceil 1 + 3.322 \log_{10}(n) \rceil$ where n is the number of observations and $\lceil \cdot \rceil$ denotes the ceiling function (round up to the nearest integer).

EXAMPLE

If a data set has 50 observations, $k = \lceil 1 + 3.322 \times 1.699 \approx 6.64 \rceil = 7$ so we would choose seven classes.

Note that the 2^k rule and Sturges' rule do not always produce the same number of classes. Either method is acceptable — what matters is that you choose one and apply it consistently throughout the rest of the frequency distribution construction.

2. Calculate the Width of Each Class: The width of a class is determined using the formula:

$$Width = \frac{Max - Min}{Number\ of\ Classes}$$

EXAMPLE

: If the data set has 50 observations and the minimum value 20 and the maximum is 78, then the width of each class is $58/6 \approx 9.7$ when using the 6 classes suggested by the 2^k rule. We can round up and use a class width of 10. It is important to note to always round up, as this ensures that all data points are included in an class.

3. Establish Class Limits: The class limits define the range of values in each class. These limits should be chosen such that each data point belongs to only one class.

EXAMPLE

Consider a data set of 50 observations where each class has a width of 10. Set the class limits of the first class to [20,30). Note that the square bracket indicates that the point should be included in the class, whereas) indicates that the point should not be included

in the class. The six classes would be $[20,30)$, $[30,40)$, $[40,50)$, $[50,60)$, $[60,70)$, and $[70,80)$. By choosing these classes, each point belongs to only one class.

Determining the Classes for the Dow Jones Industrial

The **Dow Jones Industrial Average (DJIA)** is one of the most widely followed stock market indexes in the world. It tracks the stock prices of 30 large, well-known U.S. companies — such as Apple, Microsoft, and Goldman Sachs — and summarizes their collective performance into a single number. When people say "the market was up today," they are often referring to the Dow. It serves as a barometer of the overall health of the U.S. economy and is updated continuously during trading hours.

Let's look at a snapshot of the Dow Jones Industrial 30 stock prices. Below you can see the data:

$277	$175	$202	$383	$358
$188	$294	$157	$212	$42
$149	$410	$303	$165	$203
$104	$23	$60	$287	$121
$312	$52	$54	$158	$501
$96	$95	$43	$189	$201

Let's follow the steps to build the frequency distribution.

1. Determine the Number of Classes: Here we choose five classes since $2^5 = 32$ is greater than 30 by the least amount.

2. Calculate the Width of Each Class: The smallest values in the data set is 23 and the maximum is 501. This gives us a range of 478. Now we can just take the range and divide by five to get 95.6. To make things simple we can round to 100 and use a class width of 100.

3. Establish Class Limits: Since we have rounded up we can be flexible with our class limits. The following class limits are suggested $[20,120)$, $[120,220)$, $[220,320)$, $[320,420)$, and $[420,520)$. Note that each class has a width of 100, and that each data point belongs to one single class.

2.3 Frequency Distributions and Bar Graphs in Excel (Categorical)

To construct frequency distributions in Excel, we will use pivot tables. A **pivot table** is an Excel tool that automatically summarizes and counts data, making it ideal for building frequency distributions. It allows you to quickly organize raw data into a structured table without writing any formulas.

Let's use the food data to construct a frequency distribution. Download the data below and follow the steps to build the pivot table.

Download Data

1. Select the Data: Select the data range in Excel while including the headers.

2. Insert a Pivot Table: Go to `Insert → Pivot Chart`. In the dialog box, confirm the data range and choose `Existing Worksheet` to place the pivot table next to the data. Click `OK`.

3. Configure the Pivot Table: In the PivotTable Fields panel on the right, drag the `Favorite Food` field to the `Categories` area and drag it again to the `Values` area. Excel will automatically count the occurrences of each food item, giving you the frequency distribution.

4. Add the Relative Frequency: Drag the `Favorite Food` field to the `Values` area a second time. By default, Excel will show it as a count. To convert it to a percentage, click on the field in the `Values` area and select `Value Field Settings`. Then click on the `Show Values As` tab and choose `% of Column Total` from the dropdown menu. Click `OK`. Excel will automatically display the proportion of each food item relative to the total, giving you the relative frequency distribution directly in the pivot table.

If you need more guidance please refer to the video demonstration below:

Video walkthrough: watch it in the online version of this chapter

2.4 Frequency Distributions and Histograms in Excel (Numerical)

To construct a frequency distribution for numerical data in Excel, we need an extra step compared to the categorical case. Since numerical data has no predefined categories, we must first create the bins manually using a helper column. Once the bins are in place, we can use a pivot table to count the observations in each class.

Let's use the Dow Jones stock price data to construct a frequency distribution. Download the data below and follow the steps.

Download Data

1. Create a Bin Column: In column C, add the header "Bins". In cell C2, enter the following `IFS` formula to assign each stock price to its corresponding class:

```
=IFS(B2<120,"[20,120)",B2<220,"[120,220)",B2<320,"[220,320)",
B2<420,"[320,420)",B2<=520,"[420,520)")
```

Copy the formula down for all 30 rows. Each stock price will now be assigned to one of the five classes defined in Section **??**. Press `Ctrl + ~` to verify the formulas are correct, then press `Ctrl + ~` again to return to normal view.

2. Insert a Pivot Table: Select the data. Go to `Insert → PivotTable`, confirm the data range, and select `Existing Worksheet`. Click `OK`.

3. Configure the Pivot Table: In the PivotTable Fields panel, drag the `Bins` field to the `Rows` area and drag it again to the `Values` area. Excel will count the number of stock prices in each bin, giving you the frequency distribution. Since the bin labels are text, Excel may sort them alphabetically, placing `[120,220)` before `[20,120)`. To fix this, right-click on the `[20,120)` bin in the pivot table and select `Move → Move to Beginning`. The bins should now appear in the correct numerical order.

4. Add the Cumulative Distribution: Drag the `Bins` field to the `Values` area a second time. Click on the field in the `Values` area and select `Value Field Settings`. Click on the `Show Values As` tab and choose `% Running Total In` from the dropdown menu. Make sure the base field is set to `Bins`. Click `OK`. Excel will display the cumulative percentage for each bin, showing the proportion of stock prices that fall at or below each class. You should obtain the following frequency distribution:

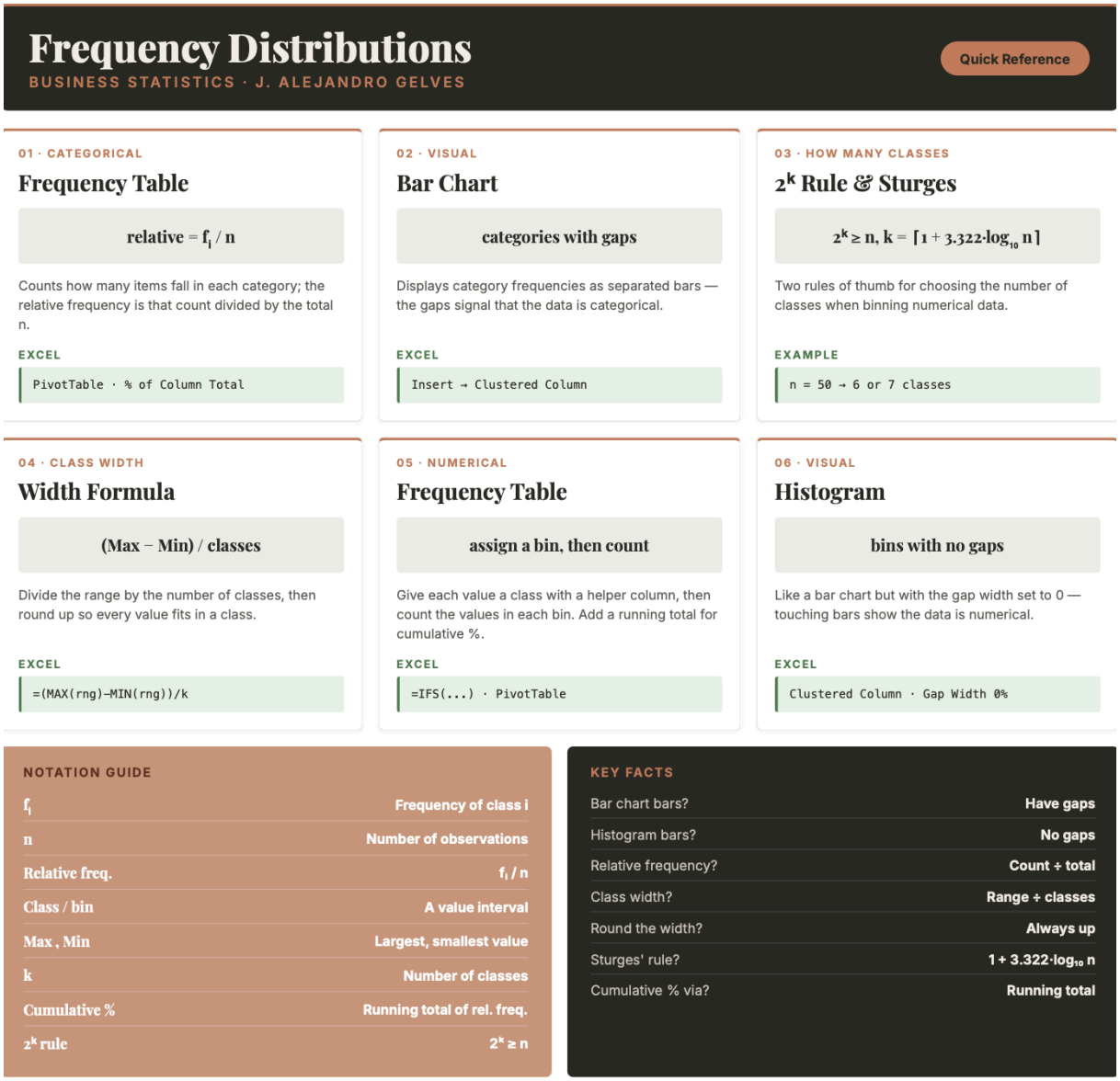
Bin	Frequency	Cumulative %
1. [20,120)	9	30.00%
2. [120,220)	12	70.00%
3. [220,320)	5	86.67%
4. [320,420)	3	96.67%
5. [420,520)	1	100.00%

5. Insert a Histogram: With the pivot table selected, go to `Insert → Charts → Clustered Column`. Excel will generate a bar chart. To convert it into a proper histogram, right-click on any bar and select `Format Data Series`. Set the `Gap Width` to `0%`. This removes the spaces between bars, which is the key visual distinction between a bar chart (used for categorical data) and a histogram (used for numerical data).

The video below walks you through each of these steps:

Video walkthrough: watch it in the online version of this chapter

2.5 Chapter Summary Cheat Sheet



2.6 Exercises

The following exercises will help you practice summarizing data with tables and simple graphs. In particular, the exercises work on:

- Developing frequency distributions for both categorical and numerical data.
- Constructing bar charts, histograms, and line charts.
- Creating contingency tables.

Answers are in the Answers to Exercises at the end of the book. Try not to peek until you have formulated your own answer and double checked your work for any mistakes.

Exercise 1

You will need the **Brain Cancer** data set to answer this question. You can download it here:

Download Data

1. Construct a frequency and relative frequency table of the *Diagnosis* variable. What was the most common diagnosis? What percentage of the sample had this diagnosis?

Answer on page ?? →

2. Construct a bar chart. Summarize the findings.

Answer on page ?? →

3. Construct a contingency table that shows the *Diagnosis* along with the *Status*. Which diagnosis had the highest number of non-survivals (0)? What was the survival rate of this diagnosis?

Answer on page ?? →

Exercise 2

You will need the **Air Quality** data set to answer this question. You can download it here:

Download Data

1. Construct a frequency distribution for the *Temp* variable. Use five classes with widths of $50 < x \leq 60$; $60 < x \leq 70$; etc. Which interval had the highest frequency? How many times was the temperature between 50 and 60 degrees?

Answer on page ?? →

2. Construct a relative frequency, cumulative frequency and the relative cumulative frequency distributions. What proportion of the time was *Temp* between 50 and 60 degrees? How many times was the *Temp* 70 degrees or less? What proportion of the time was *Temp* more than 70 degrees?

Answer on page ?? →

3. Construct the histogram. Is the distribution symmetric? If not, is it skewed to the left or right?

Answer on page ?? →

Exercise 3

You will need the **Portfolio** data set to answer this question. You can download it here:

Download Data

1. Construct a line chart that shows the returns over time for each portfolio (X and Y) by using two lines each with a unique color. Assume the data is for the period 1901 to 2000. Include also a legend that matches colors to portfolios.

Answer on page ?? →

Exercise 4

A company surveyed 40 employees about their primary department. The responses are saved in the Excel file below:

Download Data

1. Create a frequency distribution table for these departments.

Answer on page ?? →

2. Compute the relative frequencies (as percentages).

Answer on page ?? →

3. Create a bar plot of the frequencies. Which department has the highest frequency? What is its relative frequency (percentage)?

Answer on page ?? →

Exercise 5

A class of 50 students took a business statistics exam (out of 100 points). Their scores are recorded in the Excel file below:

Download Data

1. Construct a histogram using five classes with width 10, starting at 50 (bins: [50,60), [60,70), etc.). In which bin is the highest frequency? How many students scored in that range?

Answer on page ?? →

3 Descriptive Statistics III

Understanding where the "center" of a dataset lies is a fundamental step when interpreting and analyzing data. Measures of central location provide different ways to identify the "typical" value in a dataset, each with its unique strengths and limitations. Below I survey some of the most popular measures of central location. These summaries continue the work of describing our data: having seen its overall shape, we now pin down its center — a single representative value that will later anchor the estimates we carry into inference.

3.1 Sample Vs. Population Statistics

In this book you will find formulas for calculating sample statistics and population parameters. In general, you will use population formulas when you have data for the entire group you care about or when you are working with a theoretical distribution and the parameter is known; population formulas use N in denominators and estimate parameters. Use sample formulas when your data are a subset drawn from a larger population and you want unbiased estimators of the population parameters; sample formulas use n and the degrees of freedom correction. In practice most empirical work uses sample formulas because we rarely observe entire populations; below you will see both population and sample versions so you can choose the correct one depending on whether your data represent the full population or a sample.

3.2 The Mean

The **mean** is the average value for a numerical variable. It is a widely understood and straightforward measure to calculate. It incorporates all data points, providing a comprehensive representation of the data set. However, its reliance on every value also makes it sensitive to outliers or skewed distributions, which may cause it to not accurately reflect the true center of the data.

The sample statistic is estimated by:

THE MEAN (SAMPLE)

$$\bar{x} = \frac{\sum x_i}{n}$$

where Σ is the summation operator, x_i is observation i, and n is the number of observations in the sample.

The formula instructs you to add up all of the values in a variable and then divide by the sample size. If you wish to estimate the population parameter you can use:

THE MEAN (POPULATION)

$$\mu = \frac{\sum x_i}{N}$$

where N is the population size.

Let's consider a couple of example to highlight how the formula works.

EXAMPLE 1

Consider the following sample of numbers $x = \{1, 4, 2, 1\}$. The mean would be equal to $\bar{x} = \frac{1+4+2+1}{4}$ or $\bar{x} = 2$.

EXAMPLE 2

Consider the following sample of numbers $x = \{1, 4, 2, 1, 100\}$. The mean is $\bar{x} = 21.6$. Although most of the numbers are in the range $(1, 4)$, the 100 biases the mean to 21.6.

3.3 The Median

The **median** is the value in the middle when data is organized in ascending order. When n is even, the median is the average between the two middle values. The median is resistant to outliers, making it an ideal measure for skewed data or data sets with irregular distributions. It is particularly useful for ordinal data, where precise ranking is important. However, the median does not utilize all data points, which can lead to less precise comparisons compared to other measures like the mean.

EXAMPLE 1

Consider the following numbers $x = \{1, 4, 2, 1\}$. We first sort the data to obtain $x_{sorted} = \{1, 1, 2, 4\}$. Since the number of observations is even $(n = 4)$, the median is the average of the two middle numbers $(1, 2)$. $median_x = \frac{1+2}{2}$ or 1.5.

EXAMPLE 2

Consider the following numbers $x = \{1, 4, 2, 1, 100\}$. We once again sort the number obtaining $x_{sorted} = \{1, 1, 2, 4, 100\}$. Since $n = 5$ in this case we can identify the median as the third value in x_{sorted}. $median_x = 2$. Note that the inclusion of 100 in the data did not change much the measure of central location.

3.4 The Mode

The **mode** is the value with highest frequency from a set of observations. This measure is particularly useful for categorical data as it helps determine popularity of values. It can be applied to both numerical and non-numerical data sets. The mode has its limitations; it may not exist in cases where all values occur with equal frequency, and there may be multiple modes, which can complicate interpretation. Additionally, since the mode focuses only on the most frequent value, it does not account for other data points, limiting its overall utility as a comprehensive measure.

EXAMPLE 1

Consider the following numbers $x = \{1, 4, 2, 1\}$. Since 1 is repeated twice and all other numbers just repeated once, $x_{mode} = 1$.

EXAMPLE 2

Consider the following numbers $x = \{1, 4, 2, 1, 4\}$. Now 4 is also repeated twice. The variable has two modes $x_{mode} = \{1, 2\}$. x is said to be bimodal.

3.5 The Weighted Mean

The weighted mean is useful in scenarios where some data points are more significant than others, such as in financial portfolios, grade point averages, or survey results, as it accounts for variability in importance across observations. This measure requires additional information in the form of weights (w_i), which may not always be available or accurate. It is calculated the sum product of values (x_i) and weights (w_i) and then dividing by the sum of weights. Mathematically, the weighted average is:

THE WEIGHTED MEAN

$$\bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}$$

where w_i represents the weight of the ith observation.

Let's consider once more a couple of examples.

EXAMPLE 1

Consider three different stocks $S = \{T, C, X\}$ with stock returns of $R = \{2, 4, 10\}$. Each stock has a weight in the portfolio of $W = \{0.3, 0.2, 0.5\}$. The average return of the portfolio is $\bar{x}_{weighted} = \frac{0.6+0.8+5}{1}$ or $\bar{x}_{weighted} = 6.4$.

EXAMPLE 2

A company evaluates job candidates using three criteria: technical skills, communication, and teamwork. The criteria and their weights are shown in the table below:

Criteria	Weight	Score
Technical Skills	0.50	88
Communication	0.30	76
Teamwork	0.20	82

The weighted mean score for the candidate is:

$$\bar{x}_w = (0.50 \times 88) + (0.30 \times 76) + (0.20 \times 82)$$

$$= 44 + 22.8 + 16.4 = 83.2$$

The candidate's overall score is 83.2 out of 100. Note that a simple average would have given $(88 + 76 + 82)/3 = 82$, a slightly different result. The weighted mean reflects the company's priorities by giving more importance to technical skills.

3.6 The Geometric Mean

The **geometric mean** is a multiplicative average that is less sensitive to outliers relative to the arithmetic mean. It is particularly useful when averaging rates or levels that compound over time. The formula depends on whether the data represents rates or levels.

When working with rates, we add 1 to each value because rates represent changes relative to a base value. The geometric mean is given by:

THE GEOMETRIC MEAN (RATES)

$$\bar{x}_g = \sqrt[n]{(1 + r_1) \times (1 + r_2) \times \ldots \times (1 + r_n)} - 1$$

where r_i are rates of change and n is the number of periods.

When working with levels, the values already reflect standalone quantities rather than relative changes, so no adjustment is needed. The geometric mean simplifies to:

THE GEOMETRIC MEAN (LEVELS)

$$\bar{x}_g = \sqrt[n]{x_1 \times x_2 \times ... \times x_n}$$

where x_i are positive values such as prices, proportions, or index values.

To illustrate the distinction, we work through two examples below.

EXAMPLE 1

A company tracks its market share across four regions: $x = \{0.20, 0.30, 0.10, 0.10\}$. Since market share is a proportion of the whole — a level, not a rate of change — we apply the levels formula directly:

$$\bar{x}_g = \sqrt[4]{0.20 \times 0.30 \times 0.10 \times 0.10}$$
$$= \sqrt[4]{0.0006} = 0.156$$

The geometric mean market share across the four regions is 15.6%.

EXAMPLE 2

An investment fund reports the following annual returns over four years:

Year	Return	Growth Factor
2020	10%	1.10
2021	25%	1.25
2022	-8%	0.92
2023	15%	1.15

Since these are rates of return, we add 1 to each value to obtain the growth factor before applying the formula. The geometric mean is:

$$\bar{x}_g = \sqrt[4]{1.10 \times 1.25 \times 0.92 \times 1.15} - 1$$
$$= \sqrt[4]{1.4539} - 1 = 0.0977$$

The average annual return is 9.77%. Note that the arithmetic mean gives $(10 + 25 - 8 + 15)/4 = 10.5\%$, which overstates the true average. The geometric mean is the appropriate measure here because it accounts for the compounding effect of returns over time.

Date	People	Approval
01/01/24	50	25%
02/01/24	100	25%
03/01/24	30	70%
04/01/24	250	85%

3.7 Measures of Central Location in Excel

Excel has a collection of functions that calculate measures of central location. Let's consider the following data on approval ratings:

You can also download the data here:

Download Data

Mean: To calculate the mean in Excel use `=AVERAGE(C2:C5)`, where column C contains the approval ratings. The mean approval rating is 0.5125.

Median: To calculate the median in Excel use `=MEDIAN(C2:C5)`. The median approval rating is 0.475.

Mode: To calculate the mode in Excel use `=MODE(C2:C5)`. Note that `MODE` only returns one value even if there are multiple modes. For the approval data, the mode is 0.25.

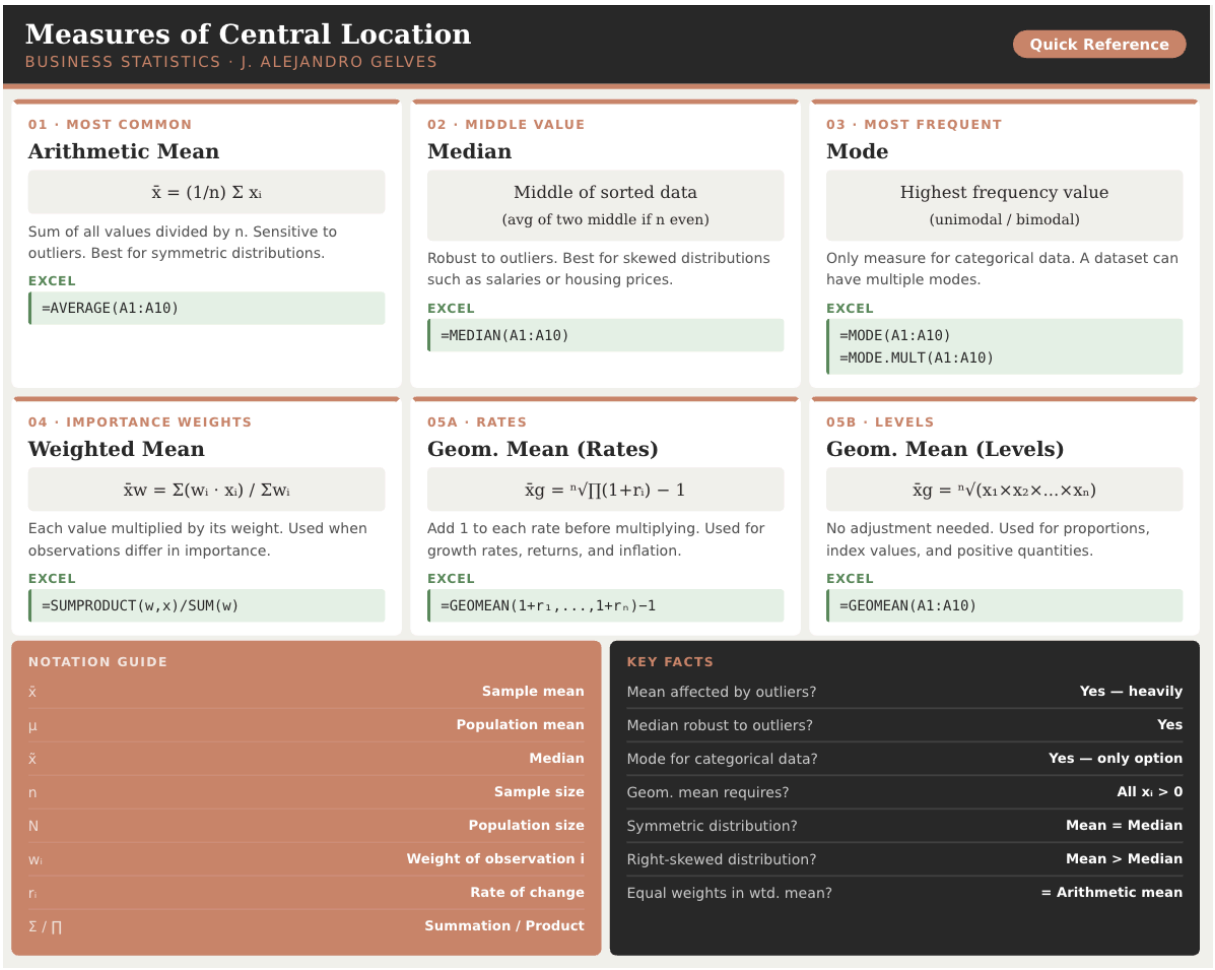
Weighted Mean: To calculate the weighted mean in Excel use `=SUMPRODUCT(B2:B5, C2:C5) / SUM(B2:B5)`, where column B contains the number of people surveyed and column C contains the approval ratings. This formula multiplies each approval rating by its corresponding weight and divides by the total number of people. The weighted mean approval rating is 0.6302326.

Geometric Mean: To calculate the geometric mean in Excel use `=GEOMEAN(C2:C5)`. Note that since approval ratings are already proportions there is no need to add 1 to each value before applying the formula. The geometric mean approval rating is 0.4391.

3.8 Excel Function Summary

- `=AVERAGE()` calculates the mean for a range of numbers.
- `=MEDIAN()` calculates the median for a range of numbers.
- `=MODE()` returns the most frequently occurring value in a range.
- `=SUMPRODUCT()` multiplies corresponding values in two ranges and returns their sum, useful for calculating the weighted mean.
- `=GEOMEAN()` calculates the geometric mean for a range of numbers.

3.9 Chapter Summary Cheat Sheet



3.10 Exercises

The following exercises will help you practice the measures of central location. In particular, the exercises work on:

- Calculating the mean, median, and the mode.
- Calculating the weighted average.
- Applying the geometric mean for growth rates and returns.

Answers are in the Answers to Exercises at the end of the book. Try not to peek until you have a formulated your own answer and double checked your work for any mistakes.

Exercise 1

For the following exercises, make your calculations by hand and verify results using Excel functions when possible.

1. Use the following observations to calculate the mean, the median, and the mode.

8	10	9	12	12

Answer on page ?? →

2. Use following observations to calculate the mean, the median, and the mode.

-4	0	-6	1	-3	-4

Answer on page ?? →

3. Use the following observations, calculate the mean, the median, and the mode.

20	15	25	20	10	15	25	20	15

Answer on page ?? →

Exercise 2

You will need the **OJ** data set to answer this question. You can get the dataset here:

Download Data

1. Find the mean price for Country Hill (*PriceCH*) and Minute Maid (*PriceMM*).

Answer on page ?? →

2. Find the mean price of Country Hill (*PriceCH*) at each store (*StoreID*). Which store provides the better price?

Answer on page ?? →

3. Find the median price paid by Country Hill (*PriceCH*) purchasers (*Purchase*) in all stores? Which store had the better median price?

Answer on page ?? →

Exercise 3

1. Over the past year an investor bought TSLA. She made these purchases on three occasions at the prices shown in the table below. Calculate the average price per share.

Date	Price Per Share	Number of Shares
February	250.34	80
April	234.59	120
Aug	270.45	50

Answer on page ?? →

2. What would have been the average price per share if the investor would have bought equal amounts of shares each month?

Answer on page ?? →

Exercise 4

1. Consider the following observations for the consumer price index (CPI). Calculate the inflation rate (Growth Rate of the CPI) for each period.

1	2	3	4	5
1.0	1.3	1.6	1.8	2.1

Answer on page ?? →

2. What is the average growth rate for the inflation rate?

Answer on page ?? →

Exercise 5

Suppose that you want to invest $1000 dollars in a stock that is predicted to yield the following returns in the next four years. Calculate both the arithmetic mean and the geometric mean. Use the geometric mean to estimate how much money you would have by the end of year 4.

Year	Annual Return
1	17.3
2	19.6
3	6.8
4	8.2

Answer on page ?? →

Exercise 6

You will need the **Hitters** dataset containing data on Major League Baseball players, including their salaries (in thousands of dollars) from 1987. You can get the dataset here:

Download Data

1. Compute the overall mean and median salary for all players. Which measure (mean or median) better represents the typical salary?

Answer on page ?? →

2. filter to only players with at least one home run (HmRun >= 1) and compute the mean and median salary for this subgroup.

Answer on page ?? →

4 Descriptive Stats IV

Measures of dispersion provide insights into how much the values in a data set deviate from the central tendency. In a business context, understanding variability is essential for assessing risk, as it helps decision-makers anticipate fluctuations in revenue, costs, and market conditions. Below, we explore the main statistics that help us quantify variability and use them to make better business decisions. Variability is central to everything that follows: it is precisely because data vary that a sample gives only an imperfect view of the world, so measuring that variation here lays the groundwork for the probability and inference chapters to come.

4.1 The Range

This is the simplest measure of dispersion, defined as the difference between the largest and smallest values in a variable. While straightforward, the range only accounts for the extremes and does not reflect the variability within the variable. The **range** is calculated by:

THE RANGE

$$Range = Maximum - Minimum$$

EXAMPLE

Consider the data $x = \{480, 1050, 1400, 2500, 3200\}$. The range is given by $Range = 3200 - 480 = 2720$.

4.2 The Variance

The variance gives us a more comprehensive look at dispersion by considering how each data point deviates from the mean. It summarizes the squared deviations from the mean by finding their average.The **variance population parameter** is given by:

THE VARIANCE (POPULATION)

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

where μ is the population mean and N is the population size.

The **variance sample statistic** is given by:

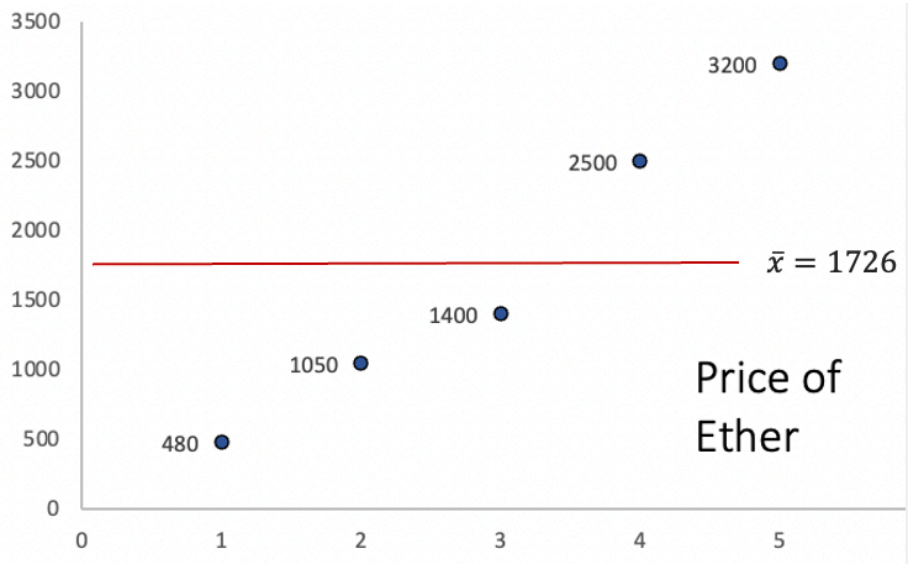
THE VARIANCE (SAMPLE)

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

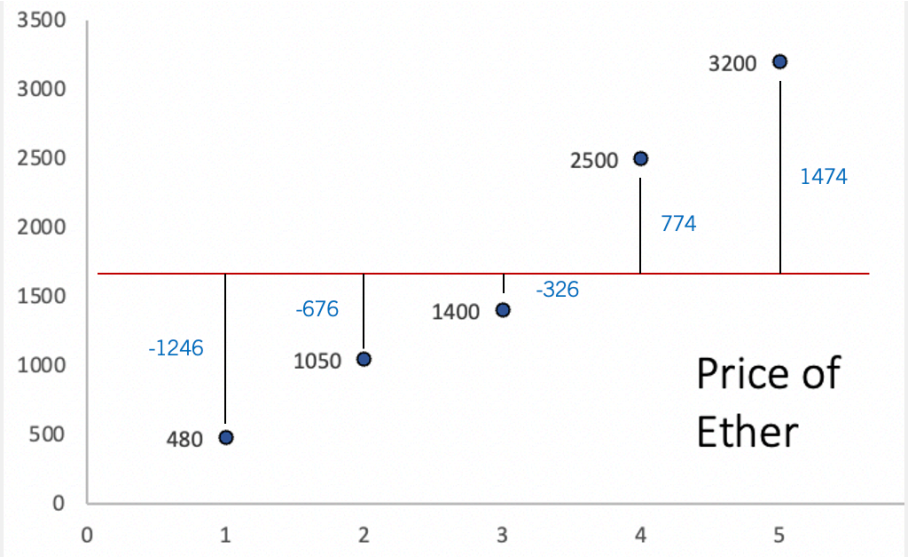
where $\bar{x}$ is the sample mean and n is the sample size.

Variance Intuition

Let's consider a sample of the price of Ether, the native cryptocurrency of the Ethereum blockchain. Ether is one of the largest cryptocurrencies by market capitalization and is widely used to pay for transactions and services on the Ethereum network. Unlike traditional assets such as stocks or bonds, cryptocurrencies like Ether are not backed by earnings or government guarantees, which makes their prices highly sensitive to speculation, regulatory news, and shifts in investor sentiment. As a result, Ether is known for dramatic price swings — sometimes gaining or losing double-digit percentages within a single day — making it an ideal case study for measures of dispersion. Below is a graph of the prices:



The y axis represents the price, while the x axis represents the period of time. The red line is the average price of Ether. The way the variance calculates dispersion, is by first finding the distance between each point and the average. The image below illustrates these distances:



Ideally, our measure of dispersion would simply average the distances from each data point to the mean, giving us a clear indication of how much the data varies around this central point. Since the mean is a measure that balances the distances from the mean, the sum of all deviations is equal to zero. This "balancing" effect is visually apparent in the graph, where positive deviations are exactly offset by negative ones. To address this issue, the variance squares each deviation from the mean before finding the average.

Variance Step By Step Calculation

Consider the data $x = \{480, 1050, 1400, 2500, 3200\}$. Treating this as a sample, let's calculate the variance step by step.

Step 1 — Calculate the sample mean:

$$\bar{x} = \frac{480 + 1050 + 1400 + 2500 + 3200}{5} = \frac{8630}{5} = 1726$$

Step 2 — Calculate the deviation of each observation from the mean:

x_i	$x_i - \bar{x}$
480	$480 - 1726 = -1246$
1050	$1050 - 1726 = -676$
1400	$1400 - 1726 = -326$
2500	$2500 - 1726 = 774$
3200	$3200 - 1726 = 1474$

Step 3 — Square each deviation:

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
480	−1246	1,552,516
1050	−676	456,976
1400	−326	106,276
2500	774	599,076
3200	1474	2,172,676

Step 4 — Sum the squared deviations:

$$\sum (x_i - \bar{x})^2 = 1{,}552{,}516 + 456{,}976 + 106{,}276 + 599{,}076 + 2{,}172{,}676 = 4{,}887{,}520$$

Step 5 — Divide by $n - 1$:

$$s^2 = \frac{4{,}887{,}520}{5 - 1} = \frac{4{,}887{,}520}{4} = 1{,}221{,}880$$

The sample variance is $s^2 = 1{,}221{,}880$. Note that the variance is expressed in squared units of the original data, which makes it difficult to interpret directly. This motivates the use of the standard deviation, which brings the measure back to the original units.

4.3 The Standard Deviation

The standard deviation is derived by taking the square root of the variance. Remember, the variance employs squared deviations to eliminate negative values and calculate spread. By taking the square root, the standard deviation reverts the measure of dispersion back to the original units of the variable. This transformation makes the standard deviation more intuitive; it directly quantifies how much each data point deviates from the mean in the same units as the variable itself. Consequently, the standard deviation is a clear and intuitive measure of variability.

To find the **population standard deviation**, take the square root of the variance. For the population parameter use:

THE STANDARD DEVIATION (POPULATION)

$$\sigma = \sqrt{\sigma^2}$$

For the **sample standard deviation** use:

THE STANDARD DEVIATION (SAMPLE)

$$s = \sqrt{s^2}$$

EXAMPLE

Consider once more the price of Ether. That is. $x = \{480, 1050, 1400, 2500, 3200\}$. In the previous section we found that the variance $s^2 = 1221880$. Using the square root we find that $s = \sqrt{1221880} = 1105.387$. If the price of Ether is in dollars, then, the price varies from the mean 1105.39 dollars on average.

4.4 Mean Absolute Deviation

Another measure of data variability is the Mean Absolute Deviation (MAD), which calculates variability using absolute deviations from the mean. This approach not only keeps the measure in the same units as the data but also makes it less sensitive to large deviations. Practically, the **Mean Absolute Deviation** measures the average deviation from the mean, by using absolute deviations. It is calculated by:

THE MEAN ABSOLUTE DEVIATION (POPULATION)

$$MAD = \frac{\sum |x_i - \mu|}{N}$$

where the | | operator indicates absolute value.

You can estimates the mad for a sample with:

THE MEAN ABSOLUTE DEVIATION (SAMPLE)

$$mad = \frac{\sum |x_i - \bar{x}|}{n}$$

Example: Let's consider once more the price of Ether. Recall, that if we calculate the deviations from the mean we obtain the second column in the table below:

Price	Deviations $(x_i - \bar{x})$	Squared Deviations $(x_i - \bar{x})^2$	Absolute Deviations $\lvert x_i - \bar{x} \rvert$
480	-1246	1552516	1246
1050	-676	456976	676
1400	-326	106276	326
2500	774	599076	774
3200	1474	2172676	1474

For the MAD, focus on the fourth column which lists the absolute deviations from the mean. Averaging these gives us the Mean Absolute Deviation (MAD), which equals $mad = 899.2$. This result states that, on average, each data point is roughly 900 dollars away from the mean. Note that the standard deviation is higher (988.73) since the squaring of deviations disproportionately amplifies larger ones, pushing the standard deviation above the MAD.

4.5 Coefficient of Variation

The **Coefficient of Variation** simplifies comparisons of variability across variables with different units or scales, by dividing the standard deviation by the mean. It is calculated by:

THE COEFFICIENT OF VARIATION (CV)

$$CV = \frac{s}{\bar{x}} \times 100$$

EXAMPLE

Consider the table below, that shows information on two different stocks:

Stock	Average Price	s
A	1	1
B	100	1

The third column shows the standard deviation of each stock. One could conclude that both stocks vary the same as they have the same standard deviation of one. Recall that the coefficient of variation considers the variable's scale by incorporating the mean into its calculation. Since one stock is centered around 1 dollar and the other around 100 dollars (second column), it is clear that they do not vary similarly percentage-wise. Stock A varies 100% from the mean, whereas Stock B only varies 1%. Hence, the coefficient of variation would identify Stock A as the more variable stock.

4.6 The Sharpe Ratio

The Sharpe Ratio measures how much excess return an investor receives for the extra volatility (risk) taken on beyond the risk-free rate. It essentially uses the same principle of normalization as the CV but adds the dimension of risk-adjusted performance. If the CV tells you the variability of returns in relation to their mean, the Sharpe Ratio tells you if that variability is worth it by comparing it against what you could safely earn without risk. In essence, while the CV highlights variability, the Sharpe Ratio motivates investment choices by rewarding higher returns per unit of risk taken.

In sum, the **Sharpe ratio** quantifies the excess return of an investment over the risk free return. It is calculated by:

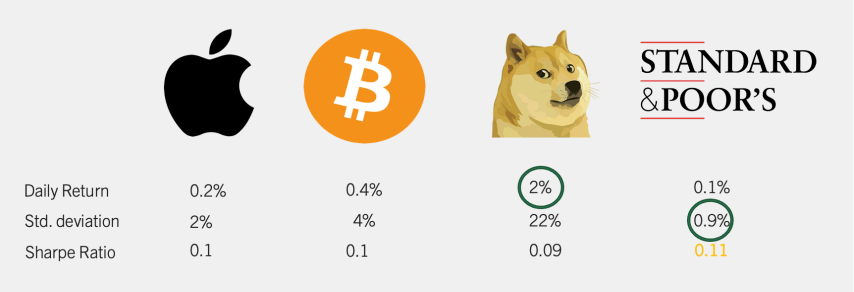
THE SHARPE RATIO

$$\frac{\bar{R_p} - R_f}{s}$$

where $\bar{R_p}$ is the mean return of the portfolio, R_f is the risk free return, and s is the standard deviation.

EXAMPLE

Consider the table below that includes a collection of investments.



The table shows four investments (Apple, Bitcoin, Shiba, and S&P). Just looking at the daily return, it is clear that Shiba provides the best returns. However, the cost for that high return is reflected in variability (22% for Shiba). The S&P is clearly the safest investment with a standard deviation of only 0.9%. The Sharpe Ratio, marries these two metrics showing the return per unit of risk taken over the free rate. If we assume a 0% risk free rate, the investment with the highest Sharpe Ratio is the S&P at 0.11.

4.7 Measures of Dispersion in Excel

Excel has a collection of functions that calculate measures of dispersion. Let's use daily stock return data for three assets — Apple (AAPL), the S&P 500 ETF (SPY), and Tesla (TSLA) — to explore these functions. The data contains daily returns for each stock from January 2020 to July 2021.

Download Data

Count: We begin by finding the number of observations for each stock using `=COUNTIF()`. Assuming the stock names are in column B and returns in column C, the number of observations for AAPL is:

Stock	N
AAPL	394
SPY	394
TSLA	394

Stock	Avg_Return
AAPL	0.1401%
SPY	0.0670%
TSLA	0.3837%

```
=COUNTIF(B:B, "AAPL")
```

The number of daily return observations for each stock is:

Average Return: To calculate the average return we use the geometric mean, since returns compound over time. Using `=GEOMEAN()` with the adjustment for rates and the particular stocks:

```
=GEOMEAN(IF(B2:B1189="AAPL", 1+C2:C1189)) - 1
```

The average daily return for each stock is:

Minimum and Maximum: To find the minimum and maximum return for each stock use `=MINIFS()` and `=MAXIFS()`. Assuming stock names are in column B and returns in column C:

```
=MINIFS(C:C, B:B, "AAPL")
=MAXIFS(C:C, B:B, "AAPL")
```

The minimum and maximum daily returns for each stock are:

Note that TSLA shows the widest range, with a minimum daily return of around -23% and a maximum of around $+18\%$, reflecting its high volatility relative to AAPL and SPY.

Variance and Standard Deviation: To calculate the variance and standard deviation use `=VARPA()` and `=STDEV.S()`. Note that Excel does not have a `=STDEVIF()` function. To calculate a conditional standard deviation use an array formula:

Stock	Min	Max
AAPL	-13.77%	11.32%
SPY	-11.59%	8.67%
TSLA	-23.65%	18.14%